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VEHICLE CONTROL DEVICE

Abstract

A vehicle control device having circuitry configured to: recognize a surrounding situation of a vehicle; recognize a driving status of the vehicle involving a driver; and issue an alarm to the driver via a predetermined alarm device based on the recognized surrounding situation and the recognized driving status. The circuitry is configured to recognize the driving status including a steering condition with respect to the vehicle and a direction of a line of sight of the driver and issue the alarm at a predetermined alarm intensity when a predetermined execution condition is satisfied based on the surrounding situation, and in a case where at least one of the steering condition and the direction of the line of sight satisfies a predetermined condition, the circuitry is configured to make the execution condition stricter and/or decrease the alarm intensity as compared with a case where the predetermined condition is not satisfied.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is based on and claims priority under 35 USC 119 from Japanese Patent Application No. 2024-032274 filed on Mar. 4, 2024, the contents of which are incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to a vehicle control device that controls a vehicle.

BACKGROUND ART

[0003] In recent years, active efforts have been made to provide access to a sustainable transportation system in consideration of vulnerable traffic participants.

[0004] As one of these efforts, research and development on driving assist techniques and automated driving techniques for vehicles such as automobiles have been made in order to further improve safety and convenience of traffic.

[0005] As an example of the driving assist technique, Japanese Patent Application Laid-Open Publication No. 2022-059939A listed below discloses a technique in which, as control related to traveling close to an edge of a lane from traveling in a center range of the lane, departure prevention control of returning the traveling close to the edge of the lane to the center of the lane is executed.

[0006] However, in the related art, there is room for improvement from the viewpoint of appropriately issuing an alarm to a driver while preventing excessive alarming that may bother the driver.

[0007] The present disclosure relates to providing a vehicle control device capable of appropriately issuing an alarm to a driver while preventing excessive alarming that may bother the driver.

SUMMARY

[0008] An aspect of the present disclosure relates to a vehicle control device for controlling a vehicle, in which the vehicle control device comprising circuitry configured to: [0009] recognize a surrounding situation of the vehicle; [0010] recognize a driving status of the vehicle involving a driver; and [0011] issue an alarm to the driver via a predetermined alarm device based on the recognized surrounding situation and the recognized driving status, in which [0012] the circuitry is configured to recognize the driving status including a steering condition with respect to the vehicle and a direction of a line of sight of the driver, [0013] the circuitry is configured to issue the alarm at a predetermined alarm intensity when a predetermined execution condition is satisfied based on the surrounding situation, and [0014] in a case where at least one of the steering condition and the direction of the line of sight satisfies a predetermined condition, the circuitry is configured to make the execution condition stricter and/or decrease the alarm intensity as compared with a case where the predetermined condition is not satisfied.

[0015] According to the present disclosure, it is possible to provide a vehicle control device capable of appropriately issuing an alarm to a driver while preventing excessive alarming that may bother the driver.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0016] Exemplary embodiments of the present disclosure will be described in detail based on the following figures, wherein:

[0017] FIG. **1** is a block diagram illustrating a schematic configuration of a vehicle including a control device according to an embodiment;

[0018] FIG. **2** is a flowchart illustrating an example of alarm control processing executed by the control device according to the embodiment; and

[0019] FIG. **3** illustrates an example of an alarm control table referred to by the control device according to the embodiment.

DESCRIPTION OF EMBODIMENTS

[0020] Hereinafter, an embodiment of a vehicle control device according to the present disclosure will be described with reference to the drawings. The following embodiment does not limit the present disclosure, and not all elements described in the following embodiment are essential to the present disclosure. Further, two or more elements described in the following embodiment may be freely combined without departing from the gist of the present disclosure. Hereinafter, the same or similar elements are denoted by the same or similar reference signs, and a description thereof may be omitted or simplified.

Vehicle

[0021] First, a vehicle according to the present embodiment will be described. A vehicle **1** according to the present embodiment illustrated in FIG. **1** (hereinafter, also referred to as a "host vehicle") is an automobile including a drive source (not illustrated), and wheels (not illustrated) including drive wheels driven by power of the drive source and steered wheels that are steerable. As an example, the vehicle **1** may be a four-wheeled automobile having a pair of left and right front wheels and a pair of left and right rear wheels.

[0022] The drive source of the vehicle **1** may be an electric motor, an internal combustion engine such as a gasoline engine or a diesel engine, or a combination of an electric motor and an internal combustion engine. The drive source of the vehicle **1** may drive the pair of left and right front wheels, the pair of left and right rear wheels, or the four wheels including the pair of left and right front wheels and the pair of left and right rear wheels. The front wheels and the rear wheels of the vehicle **1** may all be steerable steered wheels, or the front wheels or the rear wheels may be steerable steered wheels.

[0023] The vehicle **1** includes a sensor group **10**, a navigation device **20**, a control device **30** that is an example of the vehicle control device of the present disclosure, an electric power steering (EPS) system **40**, a driving force control system **50**, a braking force control system **60**, a communication unit **70**, an operation input unit **80**, and an alarm device **90**.

[0024] The sensor group **10** includes an external sensor **11** that acquires information on a periphery of the vehicle **1** (hereinafter also referred to as "peripheral information"), and a vehicle sensor **12** that acquires information on the vehicle **1** (hereinafter also referred to as "vehicle information"). Information (in other words, detection values) acquired by each sensor in the sensor group **10** is output to the control device **30**, and is used for control of the vehicle **1** (hereinafter, also referred to as "vehicle control") performed by the control device **30**.

[0025] The external sensor **11** includes, for example, a camera **111**, a sonar **112**, and a radar **113**. The camera **111** is a digital camera that images the periphery of the vehicle **1** including the front of the vehicle **1** and outputs image data of the obtained peripheral image to the control device **30**. As the camera **111**, for example, a digital camera using an imaging element such as a charge-coupled device (CCD) or a complementary metal oxide semiconductor (CMOS) can be employed.

[0026] The sonar **112** emits sound waves to the periphery of the vehicle **1** (for example, the front, the rear, and lateral sides of the vehicle **1**), and receives reflected sounds from an object present around the vehicle **1**, thereby detecting a distance to the object, an azimuth of the object, and the like. The radar **113** emits radio waves to the periphery of the vehicle **1** including the front of the vehicle **1**, and receives reflected waves from an object present around the vehicle **1**, thereby detecting a distance to the object, an azimuth of the object, and the like. As the radar **113**, for example, a millimeter wave radar can be employed.

[0027] The external sensor **11** may include light detection and ranging (LiDAR) instead of or in addition to the sonar **112** and the radar **113**. In this case, the LiDAR emits laser light to the periphery of the vehicle **1** including the front of the vehicle **1**, and receives reflected light from an object present around the vehicle **1**, thereby detecting a distance to the object, an azimuth of the object, and the like.

[0028] The vehicle sensor **12** includes, for example, a wheel sensor **121**, a vehicle speed sensor **122**, an inertial measurement unit (IMU) **123**, an occupant camera **124**, an operation detection unit **125**, and a steering touch sensor **126**.

[0029] The wheel sensor **121** detects a rotation angle of one or more wheels among the wheels of the vehicle **1**. As an example, the wheel sensor **121** detects rotation angles of a left rear wheel and a right rear wheel. As the wheel sensor **121**, for example, an angle sensor or a displacement sensor can be employed.

[0030] The vehicle speed sensor **122** detects a vehicle speed VP that is a travel speed of the vehicle **1** (in other words, a movement speed of a vehicle body). For example, the vehicle speed sensor **122** detects the vehicle speed VP based on a rotation speed of a counter shaft (not illustrated) provided in the vehicle **1**.

[0031] The inertial measurement unit **123** detects angular velocities of the vehicle **1** in a pitch direction, a roll direction, and a yaw direction, and accelerations of the vehicle **1** in a front-rear direction, a left-right direction, and an upper-lower direction. The vehicle sensor **12** may include, instead of the inertial measurement unit **123**, an acceleration sensor that detects an acceleration of the vehicle **1** in a predetermined direction and a gyro sensor that detects an angular velocity of the vehicle **1** in a predetermined direction.

[0032] The occupant camera **124** is a digital camera that images a vehicle interior of the vehicle **1** and outputs image data of the obtained vehicle interior image to the control device **30**. For example, the occupant camera **124** may be a so-called “driver monitor camera” that is provided so as to be able to image the head of an occupant from the front (in other words, image the face) who sits on the driver's seat of the vehicle **1** (hereinafter, also referred to as a “driver”). Similarly to the camera **111**, a digital camera using an imaging element such as a CCD or a CMOS can be employed as the occupant camera **124**. In the present embodiment, image data of a vehicle interior image obtained by the occupant camera **124** imaging the vehicle interior of the vehicle is information with which a direction of a line of sight of the driver can be specified.

[0033] The operation detection unit **125** detects an operation performed by using the operation input unit **80** that is provided to be operable by the driver. In the present embodiment, the operation input unit **80** can include, for example, an operation button (not illustrated) for receiving an operation to switch between on (in other words, operation) and off (in other words, non-operation) of an LKAS described later. In this case, the operation detection unit **125** can detect the operation of turning on/off the LKAS.

[0034] The steering touch sensor **126** detects whether a steering **46** of the vehicle **1** is gripped appropriately. For example, the steering touch sensor **126** is implemented by a capacitance sensor or the like. In this case, the capacitance sensor is provided at a portion touched by the driver when the steering **46** is gripped appropriately.

[0035] The navigation device **20** includes, for example, a global navigation satellite system (GNSS) receiver **21**, a touch panel **22**, and a speaker **23**. The navigation device **20** includes a storage unit (not illustrated) implemented by a flash memory or the like. The storage unit of the navigation device **20** stores a map information database (DB) **24** and the like.

[0036] The GNSS receiver **21** specifies a current position of the vehicle **1** (for example, a latitude

and a longitude of a location where the vehicle **1** is located) based on a signals received from a GNSS satellite. For example, the navigation device **20** may acquire a detection result of the vehicle sensor **12** (for example, the wheel sensor **121** or the vehicle speed sensor **122**) via the control device **30**, and specify or complement the current position of the vehicle **1** by an inertial navigation system (INS) using a detection value of the vehicle sensor **12**.
[0037] The touch panel **22** is implemented by combining a display device such as a liquid crystal display or an organic light emitting diode (OLED) with a pointing device (for example, a touch pad). The speaker **23** is configured to output a sound to an occupant (for example, the driver) of the vehicle **1**.
[0038] For example, the navigation device **20** searches for a route leading from the current position of the vehicle **1** to a destination, which is set by the driver using the touch panel **22**, by referring to the map information database **24**. Then, the navigation device **20** performs route guidance using the touch panel **22** and the speaker **23** based on a route obtained by the search. The navigation device **20** may cause the touch panel **22** to perform predetermined display in accordance with an instruction from the control device **30**. Further, the navigation device **20** may output, to the control device **30**, predetermined information such as information indicating the specified current position of the vehicle **1** or information indicating an operation received via the touch panel **22**.
[0039] The control device **30** is a computer that includes, for example, a processor configured to perform various calculations, a storage unit having a non-transitory storage medium for storing various types of information, and an input and output unit configured to control input and output of data between an inside and an outside of the control device **30** (none of which is illustrated), and executes overall control of the vehicle **1**. For example, the control device **30** is implemented by one electronic control unit (ECU) or by a plurality of ECUs working in cooperation with each other. Since specific examples of control executed by the control device **30** will be described later, the description thereof will be omitted here.
[0040] The EPS system **40** includes a steering angle sensor **41**, a torque sensor **42**, an EPS motor **43**, a resolver **44**, and an EPS ECU **45**.
[0041] The steering angle sensor **41** detects a steering angle θst of the steering **46** and outputs information indicating the detected steering angle θst to the EPS ECU **45**. The torque sensor **42** detects a steering torque TQ that is a torque applied to the steering **46** of the vehicle **1**, and outputs information indicating the detected steering torque TQ to the EPS ECU **45**.
[0042] The EPS motor **43** applies a driving force or a reaction force to a steering column **47**, which is coupled to the steering **46**, in accordance with an instruction from the EPS ECU **45**, thereby assisting the driver in operating the steering **46**. The resolver **44** detects a rotation angle θm of the EPS motor **43** and outputs information indicating the detected rotation angle θm to the EPS ECU **45**.
[0043] The EPS ECU **45** is a computer that includes, for example, a processor configured to perform various calculations, a storage unit having a non-transitory storage medium for storing various types of information, and an input and output unit configured to control input and output of data between an inside and an outside of the EPS ECU **45** (none of which is illustrated), and controls the EPS system **40** (for example, the EPS motor **43**), and the EPS ECU **45** is implemented by one or two or more ECUs. For example, the EPS ECU **45** controls the EPS system **40** (for example, the EPS motor **43**) based on the steering angle θst detected by the steering angle sensor **41**, the steering torque TQ detected by the torque sensor **42**, the rotation angle θm detected by the resolver **44**, and the like.
[0044] The EPS system **40** (for example, the EPS ECU **45**) may output information indicating the steering angle θst detected by the steering angle sensor **41**, the steering torque TQ detected by the torque sensor **42**, the rotation angle θm detected by the resolver **44**, and the like to the control device **30**. Further, the EPS system **40** (for example, the EPS ECU **45**) may output information indicating a steering speed ω of the steering **46** to the control device **30**. In this case, the steering

speed ω is obtained by, for example, differentiating the steering angle θst with respect to time.
[0045] The driving force control system **50** includes a drive ECU **51**, and is configured to control a driving force of the vehicle **1**. The drive ECU **51** is a computer that includes, for example, a processor configured to perform various calculations, a storage unit having a non-transitory storage medium for storing various types of information, and an input and output unit configured to control input and output of data between an inside and an outside of the drive ECU **51** (none of which is illustrated), and controls the driving force control system **50**, and the drive ECU **51** is implemented by one or more ECUs. For example, based on an operation on an accelerator pedal **52** provided in the vehicle **1**, the drive ECU **51** controls the power output from the drive source of the vehicle **1**. The drive ECU **51** can also control the driving force control system **50** (for example, a drive source) according to an instruction from the control device **30**.
[0046] The braking force control system **60** includes a braking ECU **61**, and is configured to control a braking force of the vehicle **1**. The braking ECU **61** is a computer that includes, for example, a processor configured to perform various calculations, a storage unit having a non-transitory storage medium for storing various types of information, and an input and output unit configured to control input and output of data between an inside and an outside of the braking ECU **61** (none of which is illustrated), and controls the braking force control system **60**, and the braking ECU **61** is implemented by one or more ECUs. For example, the braking ECU **61** controls a braking force of the vehicle **1** by controlling a brake device (not illustrated) provided in the vehicle **1** based on an operation on a brake pedal **62** provided in the vehicle **1**. Here, the brake device includes, for example, a brake caliper, a cylinder that transmits a hydraulic pressure to the brake caliper, and an electric motor that generates a hydraulic pressure in the cylinder. The braking ECU **61** controls the electric motor of the brake device such that a braking force corresponding to the operation on the brake pedal **62** is generated. The braking ECU **61** can also control the braking force control system **60** (for example, a brake device) according to an instruction from the control device **30**.
[0047] The communication unit **70** is a communication interface that communicates with an external device **2** according to the control of the control device **30**. That is, the control device **30** can communicate with the external device **2** via the communication unit **70**. Examples of the external device **2** may include a terminal device (for example, a smartphone) of the driver and a server device managed by a manufacturer of the vehicle **1**. For example, a mobile communication network such as a cellular line, Wi-Fi (registered trademark), or Bluetooth (registered trademark) can be adopted for the communication between the vehicle **1** and the external device **2**.
[0048] The alarm device **90** is a device that alarms the driver according to the control of the control device **30**. The alarm device **90** includes, for example, a multi-information display (MID) **91** and a buzzer **92**. The MID **91** is implemented by, for example, a display device such as a liquid crystal display or an OLED, and is provided at a position (for example, in a meter panel of the vehicle **1**) that the driver can visually recognize. In the present embodiment, the MID **91** can display a predetermined alarm image in accordance with an instruction from the control device **30**. The alarm image can be, for example, an image indicating that there is a possibility that the vehicle **1** departs from a host lane. Here, the host lane is a lane in which the vehicle **1** travels.
[0049] The buzzer **92** is configured to output a predetermined alarm sound. In the present embodiment, the buzzer **92** can output a predetermined alarm sound in accordance with an instruction from the control device **30**. One of the buzzer **92** and the speaker **23** may be provided to serve both roles thereof. That is, the "buzzer **92**" in the following description may be replaced with the "speaker **23**".
Control Device
[0050] Next, the control device **30** will be described in more details. The control device **30** includes, for example, a first recognition unit **31**, a second recognition unit **32**, and an alarm control unit **33** as functional units implemented by the processor executing a program stored in the storage

unit of the control device **30**.

[0051] The first recognition unit **31** recognizes a surrounding situation of the vehicle **1**. For example, the first recognition unit **31** performs sensor fusion processing on detection results obtained by some or all of the camera **111**, the sonar **112**, and the radar **113** in the external sensor **11**, and recognizes the surrounding situation of the vehicle **1** based on a processing result.

[0052] More specifically, the first recognition unit **31** recognizes a position, a type, a speed, an acceleration, and the like of an object present around the vehicle **1**. At this time, the first recognition unit **31** recognizes the position of the object as a position on absolute coordinates in which a representative point (for example, a center of gravity and a center of a drive shaft) of the vehicle **1** is set as an origin. Accordingly, a relative position between the vehicle **1** and the object present around the vehicle **1** can be recognized. In the absolute coordinates described above, the position of the object may be represented by using a representative point such as a center of gravity or a corner of the object, or may be represented as a region. Examples of objects that can be recognized by the first recognition unit **31** include traffic participants such as other vehicles and pedestrians, traveling lane boundaries such as division lines and curbs that define lanes, and road signs such as speed signs and lane type signs.

[0053] According to the first recognition unit **31**, for example, a surrounding situation including an obstacle present around the vehicle **1** can be recognized. Here, examples of the obstacle may include another traffic participant (for example, another vehicle or a pedestrian) present around the vehicle **1**, and a fallen object on a road.

[0054] The first recognition unit **31** can also recognize the surrounding situation including a shape of the host lane in which the vehicle **1** travels. For example, the first recognition unit **31** can recognize the shape of the host lane based on a traveling lane boundary recognized from a peripheral image or the like captured by the camera **111**. Here, examples of the traveling lane boundary may include a division line, a road shoulder, a curb, a separation zone, and a guard rail that define a lane.

[0055] Further, the first recognition unit **31** can also recognize a position of the vehicle **1** with respect to the host lane (for example, a distance from the vehicle **1** to the traveling lane boundary of the host lane, or a time required until the vehicle **1** reaches the traveling lane boundary of the host lane). The first recognition unit **31** may also recognize a surrounding situation including other road events such as a stop line, a traffic light, a road sign, and a toll gate of a toll road.

[0056] The second recognition unit **32** recognizes a driving status of the vehicle **1** involving the driver. For example, the second recognition unit **32** recognizes a driving status including a steering condition with respect to the vehicle **1** and a direction of a line of sight of the driver, based on a detection result obtained by one or both of the occupant camera **124** and the steering touch sensor **126** in the vehicle sensor **12**. Here, the steering condition includes, for example, one or both of the steering torque TQ applied to the steering **46** and a gripping state of the steering **46**.

[0057] For example, the second recognition unit **32** can recognize the steering torque TQ based on a detection result of the torque sensor **42**. For example, the second recognition unit **32** can recognize a gripping state of the steering **46** (in other words, whether the steering **46** is being gripped appropriately) based on a detection result obtained by one or both of the occupant camera **124** and the steering touch sensor **126**.

[0058] For example, the second recognition unit **32** can recognize the direction of the line of sight of the driver based on a vehicle interior image captured by the occupant camera **124**. At this time, the second recognition unit **32** may recognize a direction of the face of the driver as the direction of the line of sight of the driver. That is, the “direction of the line of sight of the driver” in the following description may be replaced with the “direction of the face of the driver”.

[0059] The alarm control unit **33** issues an alarm to the driver via the alarm device **90** based on the surrounding situation recognized by the first recognition unit **31** and the driving situation recognized by the second recognition unit **32**. For example, the alarm control unit **33** issues an alarm to the driver by displaying a predetermined alarm image on the MID **91** in the alarm device **90**. The alarm control unit **33** may issue an alarm to the driver by outputting a predetermined alarm sound from the buzzer **92** in the alarm device **90** instead of or in addition to the alarm image to the MID **91**.

[0060] More specifically, when a predetermined execution condition (hereinafter, also referred to as an “alarm condition”) is satisfied based on the surrounding situation recognized by the first recognition unit **31**, the alarm control unit **33** issues an alarm to the driver at a predetermined alarm intensity. In the following description, the alarm control unit **33** determines, based on the surrounding situation recognized by the first recognition unit **31** and the alarm condition, whether there is a possibility that the vehicle **1** departs from the host lane, and when it is determined that there is a possibility that the vehicle **1** departs from the host lane, the alarm control unit **33** issues an alarm to the driver.

[0061] The alarm condition is determined using, for example, a distance from the vehicle **1** to a traveling lane boundary. In this case, when the distance from the vehicle **1** to the traveling lane boundary of the host lane, which is recognized by the first recognition unit **31**, is equal to or smaller than a predetermined value determined as the alarm condition, the alarm control unit **33** determines that there is a possibility that the vehicle **1** departs from the host lane, and issues an alarm to the driver. Accordingly, when there is a possibility that the vehicle **1** departs from the host lane, it is possible to call the driver's attention based on the alarm and prevent the departure of the vehicle **1** from the host lane. Accordingly, a decrease in safety due to departure of the vehicle **1** from the host lane can be prevented, and the safety of the vehicle **1** is improved.

[0062] The alarm condition may be determined using a time required until the vehicle **1** reaches a traveling lane boundary (that is, a time to line crossing (TTLC)). A smaller value of TTLC indicates a higher possibility of the vehicle **1** approaching the traveling lane boundary, that is, a higher possibility of the vehicle **1** departing from the host lane. Therefore, in this case, when the time required until reaching the traveling lane boundary of the host lane (TTLC), which is recognized by the first recognition unit **31**, is equal to or smaller than a predetermined value determined as the alarm condition, it is sufficient for the alarm control unit **33** to determine that there is a possibility that the vehicle **1** departs from the host lane, and issue an alarm to the driver.

[0063] In some cases, the driver intentionally brings the vehicle **1** close to the traveling lane boundary of the host lane. As an example, “in-cut” in which the vehicle **1** is caused to travel close to an inner side of a curve in the host lane may be intentionally performed by the driver. When the driver intentionally brings the vehicle **1** close to the traveling lane boundary of the host lane as described, if an alarm is still issued, the driver may be bothered.

[0064] Therefore, when at least one of the steering condition with respect to the vehicle **1** and the direction of the line of sight of the driver satisfies a predetermined condition (hereinafter, also referred to as an “alarm reduction condition”), the alarm control unit **33** makes an execution condition of an alarm to the driver (that is, the alarm condition) stricter and/or decreases an alarm intensity of the alarm to the driver, as compared with a case where the alarm reduction condition is not satisfied. In other words, when at least one of the steering condition with respect to the vehicle **1** and the direction of the line of sight of the driver satisfies the alarm reduction condition, the alarm control unit **33** executes one of the following (1), (2), and (3). [0065] (1) Make the execution condition of an alarm stricter as compared with the case where the alarm reduction condition is not satisfied. [0066] (2) Decrease the alarm intensity as compared with the case where the alarm reduction condition is not satisfied. [0067] (3) Make the execution condition of an alarm stricter and decrease the alarm intensity as compared with the case where the alarm reduction condition is not satisfied.

[0068] Although a specific example will be described later, the alarm reduction condition is set in advance by the manufacturer of the vehicle **1** or the like in consideration of matters that are more likely to be established in a case of intentional driving of the driver than in a case of unintentional driving of the driver. Accordingly, it is possible to prevent an alarm having a high alarm intensity or just an alarm from being issued with respect to intentional driving of the driver. On the other hand, since an alarm having a high alarm intensity or just an alarm can be issued under a loose alarm condition in the case of unintentional driving of the driver as compared with the case of intentional driving of the driver, it is possible to call the driver's attention with the alarm. Accordingly, it is possible to appropriately issue an alarm while preventing excessive alarming that may bother the driver.

[0069] The control device **30** may further include a steering control unit **34**, for example, as a functional unit implemented by the processor executing a program stored in the storage unit of the control device **30**. In this case, the steering control unit **34** is configured to execute steering control of assisting the steering with respect to the vehicle **1** based on the surrounding situation recognized by the first recognition unit **31** such that the vehicle **1** does not depart from the host lane. Hereinafter, it is assumed that control of assisting the steering such that the vehicle **1** travels while staying near a center of the host lane (hereinafter also referred to as a "lane keep assist system (LKAS)") can be executed, as an example of the steering control. For example, the steering control unit **34** executes the LKAS based on a fact that an operation of turning on the LKAS is detected by the operation detection unit **125**. Since a detailed control procedure for implementing the LKAS is well known, a detailed description thereof is omitted here.

[0070] When the LKAS is being executed by the steering control unit **34**, even if an alarm having a high alarm intensity or just an alarm is not issued, the possibility that the vehicle **1** departs from the host lane is reduced than when the LKAS is not being executed.

[0071] Therefore, in a case where the LKAS is in operation (in other words, the LKAS is being executed by the steering control unit **34**) and at least one of the steering condition with respect to the vehicle **1** and the direction of the line of sight of the driver satisfies the alarm reduction condition, the alarm control unit **33** makes the alarm condition (that is, the execution condition of an alarm) stricter and/or decreases the alarm intensity as compared with the case where the alarm reduction condition is not satisfied.

[0072] As described, when the possibility that the vehicle **1** departs from the host lane is low even if the alarm condition is made stricter and/or the alarm intensity is reduced since a configuration in which the alarm condition can be made stricter and/or the alarm intensity can be reduced during the operation of the LKAS is adopted, it is possible to prevent an alarm having a high alarm intensity or just an alarm from being issued. Accordingly, it is possible to prevent excessive alarming that may bother the driver while preventing a decrease in safety due to departure of the vehicle **1** from the host lane.

[0073] When the vehicle **1** approaches the traveling lane boundary of the host lane even though the LKAS is in operation, there is a high possibility that the approach is caused by the driver's intentional driving. For this reason, as described above, by making the alarm condition strict and/or decreasing the alarm intensity during the operation of the LKAS, it is possible to prevent an alarm having a high alarm intensity or just an alarm from being issued with respect to the intentional driving of the driver. Accordingly, it is possible to prevent excessive alarming that may bother the driver.

[0074] The alarm reduction condition includes, for example, a condition that the line of sight of the driver is directed toward a center of the host lane or toward an inner side of a curve with respect to the center. That is, when the line of sight of the driver is directed toward the center of the host lane or toward the inner side of the curve with respect to the center, there is a possibility that the in-cut is intentionally performed. Therefore, the condition that the line of sight of the driver is directed toward a center of the host lane or toward an inner side of a curve with respect to the center is included in the alarm reduction condition. Accordingly, when the driver intentionally performs the in-cut, it is possible to prevent excessive alarming that may bother the driver.

[0075] The alarm reduction condition may include a condition that the steering torque TQ is equal to or greater than a predetermined value. For example, when the vehicle **1** approaches the traveling lane boundary of the host lane during the operation of the LKAS, a torque is applied to the steering **46** by the EPS system **40** to achieve the steering angle θst for directing the vehicle **1** to the center of the host lane. In such a case, in order to achieve the steering angle θst for directing the vehicle **1** to the traveling lane boundary of the host lane, the driver needs to apply a larger torque to the steering **46**. That is, when the vehicle **1** approaches the traveling lane boundary of the host lane during the operation of the LKAS and the steering torque TQ is equal to or greater than a predetermined value, there is a high possibility that the approach is caused by the driver's intentional driving.

[0076] Therefore, the condition that the steering torque TQ is equal to or greater than a predetermined value is included in the alarm reduction condition. Accordingly, it is possible to prevent an alarm having a high alarm intensity or just an alarm from being issued with respect to the intentional driving of the driver. Accordingly, it is possible to prevent excessive alarming that may bother the driver. Here, the predetermined value is set in advance by the manufacturer of the vehicle **1** or the like in consideration of, for example, a torque that may be applied to the steering **46** by the EPS system **40** so that the vehicle **1** travels while staying near the center of the host lane during the operation of the LKAS.

[0077] The alarm reduction condition may include a condition that the steering **46** is being gripped. That is, in a case where the steering **46** is being appropriately gripped by the driver, when the vehicle **1** is about to depart from the host lane, the driver can immediately steer the vehicle **1** to avoid the departure. Therefore, the condition that the steering **46** is being gripped is included in the alarm reduction condition. Accordingly, when it is assumed that the driver can steer to avoid a risk, it is possible to prevent excessive alarming that may bother the driver.

[0078] When a state in which the alarm reduction condition is satisfied continues for a predetermined time (for example, 3 [s]), the alarm control unit **33** may make the alarm condition stricter and/or decrease the alarm intensity. In this way, it is possible to prevent a situation where the alarm condition is made stricter and/or the alarm intensity is decreased when the alarm reduction condition is satisfied accidentally for a short time. Accordingly, the alarm to the driver can be appropriately issued, and the safety of the vehicle **1** is improved. Here, the predetermined time is set in advance by the manufacturer of the vehicle **1** or the like, for example.

[0079] As an example, when a state in which the line of sight of the driver is directed toward a center of the host lane or toward an inner side of a curve with respect to the center continues for a predetermined time, the alarm control unit **33** may make the alarm condition stricter and/or decrease the alarm intensity. As another example, when a state in which the steering torque TQ is equal to or greater than a predetermined value continues for a predetermined time, the alarm control unit **33** may make the alarm condition stricter and/or decrease the alarm intensity. Further, as another example, when a state in which the steering **46** is appropriately gripped continues for a predetermined time, the alarm control unit **33** may make the alarm condition stricter and/or decrease the alarm intensity.

[0080] In a case where a width of the host lane is smaller than a predetermined value, the vehicle **1** is more likely to approach the traveling lane boundary of the host lane (that is, the traveling lane boundary that defines the host lane) as compared with a case where the width is equal to or greater than the predetermined value. Therefore, in the case where the width of the host lane is smaller than the predetermined value, if the alarm is also issued under the same alarm conditions as in the case where the width is equal to or greater than the predetermined value, excessive warning that may bother the driver is likely to occur.

[0081] Therefore, when the width of the host lane is smaller than a predetermined value (for example, 3 [m]), the alarm control unit **33** may make the alarm condition stricter and/or decrease the alarm intensity of the alarm, as compared with the case where the width is equal to or greater than the predetermined value. As described, when the width of the host lane is smaller than a predetermined value, it is possible to prevent excessive alarming, which may bother the driver, by preventing an alarm having a high alarm intensity or just an alarm from being issued.

[0082] When an obstacle whose distance from the vehicle **1** is equal to or smaller than a threshold is recognized, the alarm control unit **33** may set the alarm condition to a normal execution condition and/or set the alarm intensity to a normal alarm intensity. Accordingly, it is possible to prevent a decrease in safety of the vehicle **1** that is caused by making the alarm condition stricter and/or decreasing the alarm intensity even though an obstacle is present around the vehicle **1**.

[0083] More specifically, for example, it is assumed that the in-cut is intentionally performed by the driver and the vehicle **1** travels close to a traveling lane boundary on an inner side of a curve in the host lane. When such intentional in-cut is performed, in the vehicle **1**, as described above, the alarm condition is made stricter or the alarm intensity is decreased. It is assumed that the vehicle **1** approaches another vehicle traveling in another lane that is present on an inner side of a curve with respect to the host lane in a state where the alarm condition is made stricter and/or in a state where the alarm intensity is decreased. In such a case, the other vehicle is recognized by the first recognition unit **31** as an obstacle whose distance from the vehicle **1** is equal to or smaller than the threshold. In such a case, the alarm control unit **33** may set the alarm condition to a normal execution condition (for example, the most relaxed alarm condition among settable alarm conditions) or set the alarm intensity to a normal alarm intensity (for example, the highest alarm intensity among settable alarm intensities).

[0084] Further, a case is considered in which reliability of a surrounding situation recognized by the first recognition unit **31** (for example, a relative position between the vehicle **1** and an object present around the vehicle **1**) may not be sufficient due to several factors such as bad weather, a bad road surface condition, or a failure of the external sensor **11**. From the viewpoint of ensuring the safety of the vehicle **1**, it is not preferable to make the alarm condition stricter or decrease the alarm intensity when the vehicle is unstable in control.

[0085] Therefore, when the reliability of the surrounding situation recognized by the first recognition unit **31** is a predetermined value and the alarm reduction condition is satisfied, the alarm control unit **33** may make the alarm condition stricter and/or decrease the alarm intensity. In other words, in a case where the reliability of the surrounding situation recognized by the first recognition unit **31** is less than the predetermined value, even if the alarm reduction condition is satisfied, the alarm control unit **33** may not make the alarm condition stricter or decrease the alarm intensity. Accordingly, it is possible to prevent a situation in which, when the vehicle **1** is unstable in control, the alarm condition is made stricter or the alarm intensity is decreased, causing a decrease in safety of the vehicle **1**. In this case, for example, the control device **30** or the like may further include a processing unit that evaluates, according to a predetermined condition, the reliability of the surrounding situation recognized by the first recognition unit **31** and transmits an evaluation result to the alarm control unit **33**.

[0086] Similarly to the above, from the viewpoint of ensuring the safety of the vehicle **1**, the alarm condition may be made stricter and/or the alarm intensity may be decreased only when the relative position between the vehicle **1** and the object around the vehicle **1** can be appropriately recognized by the first recognition unit **31** and the alarm reduction condition is satisfied.

Processing Executed by Control Device

[0087] Next, an example of alarm control processing executed by the control device **30** will be described with reference to FIGS. **2** and **3**. For example, when an ignition power supply of the vehicle **1** is turned on, the control device **30** repeatedly executes a series of processing illustrated in FIG. **2** at a predetermined cycle. At this time, for example, the control device **30** determines the alarm condition and the alarm intensity by appropriately referring to an alarm control table Tb illustrated in FIG. **3**.

[0088] As illustrated in FIG. **2**, the control device **30** first determines whether the LKAS is in operation (step S**1**). If it is determined that the LKAS is not in operation (step S**1**: NO), the control device **30** executes normal alarm control (step S**7**), and ends the series of processing illustrated in

FIG. **2**.

[0089] In the normal alarm control of step S**7**, for example, the control device **30** issues an alarm to the driver at a relatively high alarm intensity when a distance to a traveling lane boundary of a host lane is equal to or smaller than L0 (for example, 0.5 [m]) as illustrated in FIG. **3**. As an example of the alarm according to the normal alarm control (in other words, an alarm having a high alarm intensity), both display of an alarm image on the MID **91** and output of an alarm sound from the buzzer **92** can be performed.

[0090] As described above, when the LKAS is not in operation, the control device **30** performs the normal alarm control in which an alarm having a high alarm intensity can be performed under the most relaxed alarm condition, whereby a decrease in safety of the vehicle **1** can be prevented.

[0091] On the other hand, if it is determined that the LKAS is in operation (step S**1**: YES), the control device **30** determines whether the vehicle **1** is stable in control (step S**2**). For example, when the reliability of the surrounding situation recognized by the first recognition unit **31** is less than a predetermined value or the first recognition unit **31** cannot appropriately recognize a relative position between the vehicle **1** and an object around the vehicle **1**, the control device **30** determines that the vehicle **1** is unstable in control.

[0092] If it is determined that the vehicle **1** is unstable in control (step S**2**: NO), the control device **30** performs the normal alarm control in step S**7**, and ends the series of processing illustrated in FIG. **2**. As described, when the vehicle **1** is unstable in control, the control device **30** executes the normal alarm control in which an alarm having a high alarm intensity can be performed under the most relaxed alarm condition, whereby it is possible to prevent a decrease in safety of the vehicle **1** caused due to a fact that alarms are reduced even though the vehicle **1** is unstable in control.

[0093] On the other hand, if it is determined that the vehicle **1** is stable in control (step S**2**: YES), the control device **30** determines whether no obstacle is present around the vehicle **1** (step S**3**). For example, when an obstacle whose distance from the vehicle **1** is equal to or smaller than a threshold is recognized, the control device **30** determines that an obstacle is present around the vehicle **1**.

[0094] If it is determined that an obstacle is present around the vehicle **1** (step S**3**: YES), the control device **30** executes the normal alarm control in step S**7**, and ends the series of processing illustrated in FIG. **2**. As described, when an obstacle is present around the vehicle **1**, the control device **30** executes the normal alarm control, whereby it is possible to prevent a decrease in safety of the vehicle **1** due to a fact that alarms are reduced even though an obstacle is present around the vehicle **1**.

[0095] On the other hand, if it is determined that no obstacle is present around the vehicle **1** (step S**3**: NO), the control device **30** determines whether a line of sight of a driver is directed toward a center of a host lane or toward an inner side of a curve with respect to the center (step S**4**). If it is determined that the line of sight of the driver is not directed toward the center of the host lane nor toward the inner side of the curve (step S**4**: NO), the control device **30** executes first alarm reduction control (step S**8**), and ends the series of processing illustrated in FIG. **2**.

[0096] In the first alarm reduction control of step S**8**, for example, the control device **30** issues an alarm to the driver at a relatively high alarm intensity when a distance to a traveling lane boundary of the host lane is L1 (for example, L1=0.3 [m], where L1<L0) as illustrated in FIG. **3**. For example, the alarm according to the first alarm reduction control may have a high alarm intensity as the alarm according to the normal alarm control, and more specifically, both display of an alarm image on the MID **91** and output of an alarm sound from the buzzer **92** can be performed. That is, in the first alarm reduction control, it is determined whether to perform alarming using an alarm condition that is stricter than in the normal alarm control, and when it is determined to perform alarming, an alarm having a high alarm intensity is issued.

[0097] As described, when the line of sight of the driver is not directed toward the center of the host lane nor toward the inner side of the curve (for example, when the driver performs in-cut unintentionally), the control device **30** executes the first alarm reduction control. Thus, it is possible to prevent the alarm condition from being made excessively strict or the alarm intensity from being excessively decreased as compared with a case where second alarm reduction control or third alarm reduction control described later is executed.

[0098] If it is determined that the line of sight of the driver is directed toward the center of the host lane or toward the inner side of the curve with respect to the center (step S**4**: YES), the control device **30** determines whether the steering **46** is being gripped appropriately and the steering torque TQ is equal to or greater than a predetermined value (step S**5**).

[0099] If it is determined that the steering **46** is not being gripped or the steering torque TQ is smaller than the predetermined value (step S**5**: NO), the control device **30** executes the second alarm reduction control (step S**9**).

[0100] In the second alarm reduction control of step S**9**, for example, the control device **30** issues an alarm to the driver at a relatively low alarm intensity when a distance to a traveling lane boundary of the host lane is equal to or smaller than L1 as illustrated in FIG. **3**. As an example of the alarm according to the second alarm reduction control (in other words, an alarm having a low alarm intensity), only one of display of an alarm image on the MID **91** and output of an alarm sound from the buzzer **92** (for example, only the display of an alarm image on the MID **91**) can be performed. That is, in the second alarm reduction control, it is determined, as in the first alarm reduction control, whether to perform alarming using an alarm condition that is stricter than in the normal alarm control, and when it is determined to perform alarming, an alarm having a low alarm intensity is issued.

[0101] On the other hand, if it is determined that the steering **46** is being gripped appropriately and the steering torque TQ is equal to or greater than the predetermined value (step S**5**: YES), the control device **30** determines whether a width of the host lane is smaller than a predetermined value (step S**6**).

[0102] If it is determined that the width of the host lane is equal to or greater than the predetermined value (step S**6**: NO), the control device **30** executes the third alarm reduction control (step S**10**), and ends the series of processing illustrated in FIG. **2**. In the third alarm reduction control of step S**10**, for example, the control device **30** issues an alarm to the driver at a relatively low alarm intensity when a distance to a traveling lane boundary of the host lane is L2 (for example, L2=0.2 [m], where L2<L1) as illustrated in FIG. **3**. For example, the alarm according to the third alarm reduction control may have a high alarm intensity as the alarm according to the second alarm reduction control, and more specifically, only one of display of an alarm image on the MID **91** and output of an alarm sound from the buzzer **92** can be performed. That is, in the third alarm reduction control, it is determined whether to perform alarming using an alarm condition that is further stricter than in the first alarm reduction control and the second alarm reduction control, and when it is determined to perform alarming, an alarm having a low alarm intensity is issued.

[0103] On the other hand, if it is determined that the width of the host lane is smaller than the predetermined value (step S**6**: YES), the control device **30** executes fourth alarm reduction control (step S**11**), and ends the series of processing illustrated in FIG. **2**. In the fourth alarm reduction control of step S**11**, for example, the control device **30** does not issue an alarm to the driver regardless of the distance to the traveling lane boundary of the host lane as illustrated in FIG. **3**.

[0104] As described above, according to the control device **30**, it is possible to appropriately issue an alarm to the driver while preventing excessive alarming that may bother the driver.

[0105] Although an embodiment of the present disclosure has been described above with reference to the drawings, it goes without saying that the present disclosure is not limited to the embodiment described above. It is apparent that those skilled in the art can conceive of various modifications and changes within the scope described in the claims, and it is understood that such modifications and changes naturally fall within the technical scope of the present disclosure.

[0106] For example, the control device **30** may store information indicating a curvature of a curve, a vehicle speed VP, a width, a departure amount, and the like in a case where a departure from a host lane of the vehicle **1** occurs, and may determine whether to perform an alarm using the stored information. As an example, the control device **30** may predict a departure amount from a host lane this time based on a curvature of a curve, a vehicle speed VP, a width, and a departure amount in a case where the vehicle **1** departed from a host lane in the past, and a current curvature of a curve, a current vehicle speed VP, and a current width. Then, the control device **30** may determine whether a distance between the vehicle **1** and a line, which is expanded outward from a traveling lane boundary of the current host lane by the predicted departure amount, is equal to or smaller than a predetermined value defined as an alarm condition, and may issue an alarm based on a determination result.

[0107] In the embodiment described above, the control device **30** determines whether the vehicle **1** is stable in control (step S**2**), and if it is determined that the vehicle **1** is stable in control (step S**2**: YES), the control device **30** can execute various types of alarm reduction control such as the first alarm reduction control, the second alarm reduction control, and the third alarm reduction control, and the present disclosure is not limited thereto. For example, the control device **30** may execute various types of alarm reduction control without executing the processing of step S**2**.

[0108] In the present specification, at least the following matters are described. Although corresponding constituent elements in the embodiment described above are shown in parentheses, the present disclosure is not limited thereto.

[0109] (1) A vehicle control device (control device **30**) for controlling a vehicle (vehicle **1**), the vehicle control device including: [0110] a first recognition unit (first recognition unit **31**) configured to recognize a surrounding situation of the vehicle; [0111] a second recognition unit (second recognition unit **32**) configured to recognize a driving status of the vehicle involving a driver; and [0112] an alarm control unit (alarm control unit **33**) configured to issue an alarm to the driver via a predetermined alarm device (alarm device **90**) based on a surrounding situation recognized by the first recognition unit and a driving status recognized by the second recognition unit, in which [0113] the second recognition unit recognizes the driving status including a steering condition with respect to the vehicle and a direction of a line of sight of the driver, [0114] the alarm control unit is capable of issuing the alarm at a predetermined alarm intensity when a predetermined execution condition is satisfied based on the surrounding situation, and [0115] in a case where at least one of the steering condition and the direction of the line of sight satisfies a predetermined condition, the alarm control unit makes the execution condition stricter and/or decreases the alarm intensity as compared with a case where the predetermined condition is not satisfied.

[0116] According to (1), when at least one of the steering condition with respect to the vehicle and the direction of the line of sight of the driver satisfies the predetermined condition, the execution condition of the alarm based on the surrounding situation of the vehicle can be made stricter, and/or the alarm intensity can be decreased. Accordingly, it is possible to appropriately issue an alarm to the driver while preventing excessive alarming that may bother the driver, and the safety of the vehicle is improved. In addition, it is possible to improve traffic safety and contribute to development of a sustainable transportation system.

[0117] (2) The vehicle control device according to (1), in which [0118] the first recognition unit recognizes the surrounding situation including a shape of a lane in which the vehicle travels, [0119] the vehicle control device further includes a steering control unit (steering control unit **34**) configured to perform steering control of assisting steering of the vehicle based on the surrounding situation such that the vehicle does not depart from the lane, and [0120] the alarm control unit makes the execution condition stricter and/or decreases the alarm intensity when the predetermined condition is satisfied during operation of the steering control.

[0121] According to (2), when the steering control of assisting the steering of the vehicle such that the vehicle does not depart from the lane is in operation, the execution condition of the alarm can be made stricter, and/or the alarm intensity can be decreased. Accordingly, when the safety of the

vehicle is unlikely to decrease even if the execution condition of the alarm is made stricter and/or the alarm intensity is decreased, the execution condition of the alarm can be made stricter and/or the alarm intensity can be decreased. Accordingly, it is possible to prevent excessive alarming that may bother the driver while preventing a decrease in safety of the vehicle.
[0122] (3) The vehicle control device according to (2), in which [0123] the predetermined condition includes a condition that the line of sight is directed toward a center of the lane or toward an inner side of a curve with respect to the center.
[0124] When the line of sight of the driver is directed toward the center of the lane in which the vehicle travels or toward the inner side of the curve with respect to the center, there is a possibility that the driver is intentionally performing in-cut. According to (3), it is possible to prevent excessive alarming, which may bother the driver, when such intentional in-cut is performed.
[0125] (4) The vehicle control device according to (2), in which [0126] the second recognition unit recognizes the steering condition including a steering torque (steering torque TQ) generated in a steering (steering **46**) of the vehicle, and [0127] the predetermined condition includes a condition that the steering torque is equal to or greater than a predetermined value.
[0128] When the steering torque is equal to or greater than the predetermined value, there is a possibility that the driver is intentionally performing in-cut. According to (4), it is possible to prevent excessive alarming, which may bother the driver, when such intentional in-cut is performed.
[0129] (5) The vehicle control device according to (2), in which [0130] the second recognition unit recognizes the steering condition including a gripping state of a steering (steering **46**) of the vehicle, and [0131] the predetermined condition includes a condition that the steering is being gripped.
[0132] In a case where the steering is being appropriately gripped by the driver, when the vehicle is about to depart from the lane, the driver can immediately steer the vehicle to avoid the departure. According to (5), when it is assumed that the driver can steer to avoid a risk, it is possible to prevent excessive alarming that may bother the driver.
[0133] (6) The vehicle control device according to any one of (1) to (5), in which [0134] the alarm control unit makes the execution condition stricter and/or decreases the alarm intensity when a state in which the predetermined condition is satisfied continues for a predetermined time.
[0135] According to (6), it is possible to prevent a situation where the execution condition of the alarm is made stricter and/or the alarm intensity is decreased when the predetermined condition is satisfied accidentally. Accordingly, the alarm to the driver can be appropriately issued, and the safety of the vehicle is improved.
[0136] (7) The vehicle control device according to any one of (2) to (5), in which [0137] in a case where a width of the lane in which the vehicle travels is smaller than a predetermined value, the alarm control unit makes the execution condition stricter and/or decreases the alarm intensity as compared with a case where the width is equal to or greater than the predetermined value.
[0138] In the case where the width of the lane in which the vehicle travels is smaller than the predetermined value, the vehicle is more likely to approach a travelling lane boundary that defines the lane, as compared with the case where the width is equal to or greater than the predetermined value. Therefore, in the case where the width of the lane is smaller than the predetermined value, if the alarm is also issued under the same alarm conditions as in the case where the width is equal to or greater than the predetermined value, excessive warning that may bother the driver is likely to occur. According to (7), in the case where the width of the lane in which the vehicle travels is smaller than the predetermined value, it is possible to prevent excessive alarming, which may bother the driver, by making the execution condition of the alarm stricter and/or decreasing the alarm intensity.
[0139] (8) The vehicle control device according to any one of (1) to (7), in which [0140] when reliability of the surrounding situation recognized by the first recognition unit is a predetermined

value and satisfies the predetermined condition, the alarm control unit makes the execution condition stricter and/or decreases the alarm intensity.

[0141] A case is also considered in which the reliability of the surrounding situation recognized by the first recognition unit is not sufficient due to several factors. From the viewpoint of ensuring the safety of the vehicle, it is not preferable to make the alarm condition stricter or decrease the alarm intensity when the vehicle is unstable in control. According to (8), it is possible to prevent a situation in which, when the vehicle is unstable in control, the alarm condition is made stricter or the alarm intensity is decreased, causing a decrease in safety of the vehicle.

[0142] (9) The vehicle control device according to any one of (1) to (8), in which [0143] the first recognition unit recognizes the surrounding situation including an obstacle present around the vehicle, and [0144] the alarm control unit sets the execution condition to a normal execution condition and/or sets the alarm intensity to a normal alarm intensity when the obstacle whose distance from the vehicle is equal to or smaller than a threshold is recognized.

[0145] According to (9), when an obstacle is present around the vehicle, the execution condition of the alarm is set to the normal execution condition and/or the alarm intensity is set to the normal alarm intensity, whereby it is possible to prevent a decrease in safety of the vehicle due to a fact that the execution condition of the alarm is made stricter and/or the alarm intensity is decreased even though the obstacle is present around the vehicle.

Claims

1. A vehicle control device for controlling a vehicle, wherein the vehicle control device comprising circuitry configured to: recognize a surrounding situation of the vehicle; recognize a driving status of the vehicle involving a driver; and issue an alarm to the driver via a predetermined alarm device based on the recognized surrounding situation and the recognized driving status, wherein the circuitry is configured to recognize the driving status including a steering condition with respect to the vehicle and a direction of a line of sight of the driver, the circuitry is configured to issue the alarm at a predetermined alarm intensity when a predetermined execution condition is satisfied based on the surrounding situation, and in a case where at least one of the steering condition and the direction of the line of sight satisfies a predetermined condition, the circuitry is configured to make the execution condition stricter and/or decrease the alarm intensity as compared with a case where the predetermined condition is not satisfied.

2. The vehicle control device according to claim 1, wherein the circuitry is configured to recognize the surrounding situation including a shape of a lane in which the vehicle travels, the circuitry is further configured to perform steering control of assisting steering of the vehicle based on the surrounding situation such that the vehicle does not depart from the lane, and the circuitry is configured to: issue the alarm when determining, based on the surrounding situation and the execution condition, that the vehicle has a possibility to depart from the lane; and make the execution condition stricter and/or decrease the alarm intensity when the predetermined condition is satisfied during operation of the steering control.

3. The vehicle control device according to claim 2, wherein the predetermined condition includes a condition that the line of sight is directed toward a center of the lane or toward an inner side of a curve with respect to the center.

4. The vehicle control device according to claim 2, wherein the circuitry is configured to recognize the steering condition including a steering torque generated in a steering of the vehicle, and the predetermined condition includes a condition that the steering torque is equal to or greater than a predetermined value.

5. The vehicle control device according to claim 2, wherein the circuitry is configured to recognize the steering condition including a gripping state of a steering of the vehicle, and the predetermined condition includes a condition that the steering is being gripped.

6. The vehicle control device according to claim 1, wherein the circuitry is configured to make the execution condition stricter and/or decrease the alarm intensity when a state in which the predetermined condition is satisfied continues for a predetermined time.
7. The vehicle control device according to claim 2, wherein in a case where a width of the lane in which the vehicle travels is smaller than a predetermined value, the circuitry is configured to make the execution condition stricter and/or decrease the alarm intensity as compared with a case where the width is equal to or greater than the predetermined value.
8. The vehicle control device according to claim 1, wherein when reliability of the recognized surrounding situation is a predetermined value and satisfies the predetermined condition, the circuitry is configured to make the execution condition stricter and/or decreases the alarm intensity.
9. The vehicle control device according to claim 1, wherein the first recognition unit recognizes the surrounding situation including an obstacle present around the vehicle, and the alarm control unit sets the execution condition to a normal execution condition and/or sets the alarm intensity to a normal alarm intensity when the obstacle whose distance from the vehicle is equal to or less than a threshold is recognized.
